ANDROID PROGRAMMING PRACTICAL 02

Module 1 · Programming Resources: colors, themes, strings, dimens and drawables

Aim

Demonstrate Android resource files and apply colors.xml, themes.xml, strings.xml, dimens.xml, and drawable resources in UI components.

Project name: ProgrammingResourcesDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an Empty Views Activity project and inspect app/src/main/res.

2. Add named colors in res/values/colors.xml and refer to them as @color/resource_name.

3. Add reusable text in strings.xml and use @string/resource_name rather than hard-coded labels.

4. Create dimens.xml entries for spacing, padding, and text size; reference them using @dimen/resource_name.

5. Add an image or drawable resource under res/drawable using a lowercase filename.

6. Define or update a theme in themes.xml and set it on the application or activity in AndroidManifest.xml.

7. Build and run the app; navigate through the sample screens and verify each resource is applied.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The app displays UI components using reusable color, string, dimension, theme, and drawable resources.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.